HYPER-3

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Author Note

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Abstract

The trial investigates which of three daily doses should go into phase III and whether any of them harms the people taking it. The effect of arm on change is identified via the backdoor route. Four hundred adults with elevated or stage-1 hypertension were randomized to standard of care or to one of three daily doses, stratified by age band. Seated systolic pressure was measured weekly, averaging readings inside weeks 5 to 8 for safety and weeks 9 to 12 for the primary endpoint. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure and for age band when pooled. The pooled 40 mg versus control contrast is -1.53 mmHg with a 90% WALD interval of -2.97 to -0.10 mmHg. The 20 mg versus control pooled contrast is -4.80 mmHg with a 90% WALD interval of -6.17 to -3.44 mmHg. The 10 mg versus control pooled contrast is -5.27 mmHg with a 90% WALD interval of -6.63 to -3.91 mmHg. For age 51 and older, the 40 mg versus control contrast is 6.20 mmHg with a 90% WALD interval of 3.98 to 8.42 mmHg. For ages 25 to 35, the 40 mg versus control contrast is -5.26 mmHg with a 90% WALD interval of -8.17 to -2.36 mmHg. Dropout is assumed to be ignorable given baseline, and the pooled contrast is assumed to answer the question.

Reported quantities: 40 mg vs control, pooled; 20 mg vs control, pooled; 10 mg vs control, pooled; 40 mg vs control, age 51+

Introduction

This paper evaluates which of three daily doses should go into phase III, and whether any of them is harming the people taking it. The inquiry centers on the estimated contrasts between the daily doses and the control group. In particular, the decision turns on the pooled contrast comparing 40 mg against control. The reference threshold for evaluating this contrast is 0.00 mmHg. An interval lying wholly on one side of 0.00 mmHg settles the question. An interval that spans 0.00 mmHg indicates that the question remains unsettled. The study also assesses whether any dose inflicts harm across the monitored comparisons. The validity of the resulting conclusions depends directly on underlying conditions. Specifically, two unresolved assumptions carry the analysis throughout. The first assumption is that dropout is ignorable given baseline. The second assumption is that the pooled contrast answers the question. Both assumptions are stated in the methods and revisited in the discussion of limitations.

Methods

The study evaluates which of three daily doses should advance to phase III and whether any dose causes harm. The target of inference is the effect of treatment arm on change in blood pressure, identified via the backdoor route. Four hundred adults with elevated or stage-1 hypertension were randomized 2:1:1:1 to standard of care or to one of three daily doses, stratified by age band. To determine which dose to take forward, the design estimates three contrasts against a shared control. Seated systolic pressure was measured weekly. Both endpoints average the readings inside a window—weeks 5–8 for safety and weeks 9–12 for the primary endpoint—rather than using a single visit. Averaging correlated readings reduces endpoint variance, allowing safety boundaries to operate within strata. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure, and for age band when the contrast is pooled across bands. Randomization licenses the intent-to-treat contrast, and adjusting for pre-randomization covariates adds precision without compromising identification. Twelve harm contrasts, comparing each dose against control overall and inside each age band, were evaluated at scheduled safety reviews against a posterior-probability boundary with a 2 mmHg clinical margin. Pre-specifying the age bands as monitored contrasts allows subgroup-specific harms to be detected before the trial ends. The analysis relies on the assumptions that dropout is ignorable given baseline and that the pooled contrast answers the trial question.

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
dropout is ignorable given baseline	units who left the study differ from those who stayed only through what was measured before randomization	is assumed and has not been checked	a tipping-point analysis over plausible departures from ignorability
the pooled contrast answers the question	a single number over all age bands describes the effect the decision is about	is known to fail	heterogeneity across the pre-specified strata

Table 1. Standing assumptions

assumption Adherence was not adjusted for: it lies on the path from arm to outcome, and conditioning on it would break the randomization.

assumes *no_conditioning_on_post_randomization_variables* (satisfied): the analysis conditions on nothing measured after assignment; adherence is a mediator and dropout is a descendant of the outcome

assumption The pooled 40 mg contrast is reported alongside the per-band contrasts, never instead of them.

assumes *the_pooled_contrast_answers_the_question* (violated): a single number over all age bands describes the effect the decision is about

Assumption ledger

Results

The pooled contrast for 40 mg versus control is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). For the stratum of participants aged 51 and older, the contrast for 40 mg versus control is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg).

40 MG VS CONTROL, POOLED	20 MG VS CONTROL, POOLED	10 MG VS CONTROL, POOLED	40 MG VS CONTROL, AGE 51+	40 MG VS CONTROL, AGE 25-35
-1.53 mmHg	-4.80 mmHg	-5.27 mmHg	6.20 mmHg	-5.26 mmHg
[-2.965, -0.1035] (90% WALD)	[-6.171, -3.437] (90% WALD)	[-6.631, -3.907] (90% WALD)	[3.985, 8.424] (90% WALD)	[-8.166, -2.363] (90% WALD)

Model checking

Model checking evaluates study completion and the fit of the estimation structure rather than estimating treatment contrasts. The diagnostic record tracks unit participation across the randomized age strata. The proportion of units still on study at the final week is 0.797. There are 372 units contributing to the primary endpoint. By age stratum, 100 units were randomized in the band aged 25 to 35, 160 units in the band aged 36 to 50, and 140 units in the band aged 51 plus. These diagnostics describe the observed data patterns rather than the effect itself. Diagnostic checks can only reduce confidence in the primary estimates when discrepancies emerge, never establish them. The model fit depends on assumptions that remain unresolved in the analysis. Specifically, the analysis relies on the unresolved assumption that dropout is ignorable given baseline. It also leaves unresolved the assumption that the pooled contrast answers the question. The diagnostic values do not settle whether these missing-data or pooling assumptions hold. They outline the completeness of the sample while leaving the underlying identification assumptions open.

UNITS STILL ON STUDY AT THE FINAL WEEK	UNITS CONTRIBUTING TO THE PRIMARY ENDPOINT	UNITS RANDOMIZED IN BAND AGE 25 35	UNITS RANDOMIZED IN BAND AGE 36 50	UNITS RANDOMIZED IN BAND AGE 51 PLUS
0.797	372	100	160	140

Discussion

The target effect is identified via the backdoor route, which allows the estimates to be interpreted as effects of treatment rather than associations, subject to the trial assumptions. For the pooled comparison of 40 mg versus control, the effect is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg), which lies entirely below the decision threshold on the favorable side. This pooled figure averages an age stratum where the dose helps with one where it harms. For the pooled comparison of 20 mg versus control, the effect is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg), falling entirely on the favorable side of the threshold. Similarly, the pooled contrast for 10 mg versus control is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg), which also lies entirely below the decision threshold. Stratum-level estimates clarify the performance of the highest dose across age groups. For 40 mg versus control among participants aged 51 and older, the effect is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), falling entirely on the unfavorable side of the threshold. This subgroup contrast reflects the harm that the safety boundary was designed to monitor, which the pooled estimate conceals. For 40 mg versus control among participants aged 25 to 35, the effect is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg), lying entirely below the decision threshold on the favorable side. These findings remain conditional on the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

Conclusions

Determining which daily dose should advance to phase III and whether any dose produces harm cannot be answered with a single summary metric, conditional on the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question. For 40 mg vs control in the age 51+ band, the estimate is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), which sits on the unfavourable side of the threshold. Four other findings sit on the favourable side of the threshold. The pooled estimate for 40 mg vs control is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). For 40 mg vs control in the age 25-35 band, the estimate is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg). For 20 mg vs control, the pooled estimate is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). For 10 mg vs control, the pooled estimate is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). Any single summary estimate averages harm with benefit across strata. When an overall average is quoted, it conceals stratum-specific harm behind overall benefit. Consequently, the disaggregated findings are the result, and the pooled figure is not a substitute for them. The target effects are identified via the backdoor route. All findings hold under the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

Limitations

Two unresolved assumptions underlie the analysis. The first assumption is that dropout is ignorable given baseline. Under this assumption, units who left the study differ from those who stayed only through what was measured before randomization. This licenses estimating treatment contrasts without modeling unobserved mechanisms of attrition. This condition is assumed and has not been checked. It would be challenged by a tipping-point analysis over plausible departures from ignorability. The second assumption is that the pooled contrast answers the question. Under this assumption, a single number over all age bands describes the effect the decision is about. This licenses evaluating outcomes using aggregate estimates across all strata. This assumption is known to fail. It is challenged by heterogeneity across the pre-specified strata.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
40 mg vs control, pooled	pooled_40	-1.53 mmHg (90% WALD -2.97 to -0.10 mmHg)	identify.ols (ANCOVA)
20 mg vs control, pooled	pooled_20	-4.80 mmHg (90% WALD -6.17 to -3.44 mmHg)	identify.ols (ANCOVA)
10 mg vs control, pooled	pooled_10	-5.27 mmHg (90% WALD -6.63 to -3.91 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 51+	oldest_40	6.20 mmHg (90% WALD 3.98 to 8.42 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 25-35	youngest_40	-5.26 mmHg (90% WALD -8.17 to -2.36 mmHg)	identify.ols (ANCOVA)
Units still on study at the final week	retention	0.797	—
Units contributing to the primary endpoint	analysed	372	—
Units randomized in band age 25 35	n_age_25_35	100	—
Units randomized in band age 36 50	n_age_36_50	160	—
Units randomized in band age 51 plus	n_age_51_plus	140	—

Table 2. Quantities and their sources

Field	Value
endpoint	mean change, weeks 9-12

interval_mass	90%
seed	20260821
world	nbs/case-studies/hypertension/hyper3.py
evidence hash	962d1a7970f3f2dfdc0297e7d9d6d16dd568a5441fc1c faa7c44837d40f9c2e5
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 3. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	route	backdoor
Identification	status	identified
Protocol	allocation	2:1:1:1
Protocol	arms	standard_of_care, dose_10, dose_20, dose_40
Protocol	enrollment weeks	16
Protocol	follow-up weeks	24
Protocol	primary endpoint week	12
Protocol	strata	age_25_35, age_36_50, age_51_plus
Protocol	units randomized	400
Measurement	minimum visits in window	3
Measurement	retention at final week	79.8%
Analysis	estimator	ANCOVA (OLS on change, baseline-adjusted)
Analysis	interval mass	90%
Analysis	pooled covariates	baseline_centred, is_age_36_50, is_age_51_plus
Analysis	within-stratum covariates	baseline_centred
Safety monitoring	clinical margin (mmHg)	2
Safety monitoring	monitored contrasts	12

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
40 mg vs control, pooled	-1.53	mmHg	-2.97 to -0.10 (90% WALD)	0.00	below threshold
20 mg vs control, pooled	-4.80	mmHg	-6.17 to -3.44 (90% WALD)	0.00	below threshold
10 mg vs control, pooled	-5.27	mmHg	-6.63 to -3.91 (90% WALD)	0.00	below threshold
40 mg vs control, age 51+	6.20	mmHg	3.98 to 8.42 (90% WALD)	0.00	above threshold
40 mg vs control, age 25-35	-5.26	mmHg	-8.17 to -2.36 (90% WALD)	0.00	below threshold

Table 3. Recorded diagnostics.

Check	Value	Unit	Interval
Units still on study at the final week	0.797	—	—
Units contributing to the primary endpoint	372	—	—
Units randomized in band age 25 35	100	—	—
Units randomized in band age 36 50	160	—	—
Units randomized in band age 51 plus	140	—	—

Figures

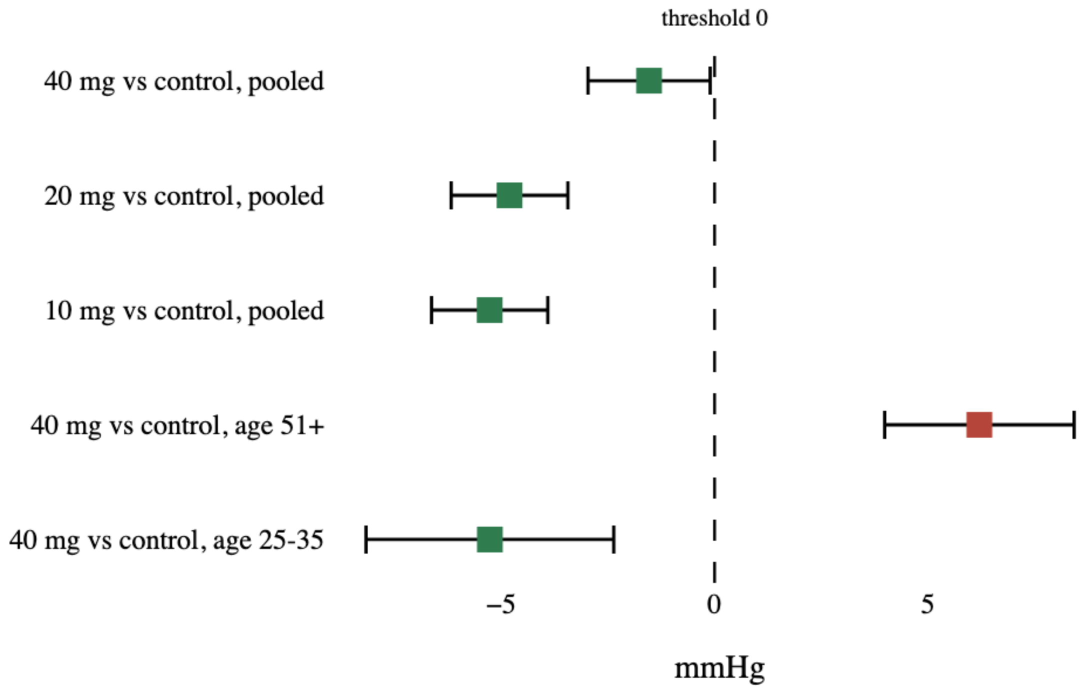

Figure 1. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.